Kazusa Tsubota

Lead ML Architect | Google Cloud & MLOps Platforms

kazusa931209@gmail.com | +817020172888 | Osaka, Japan | https://portfolio-lake-seven-63.vercel.app/ | https://www.linkedin.com/in/kazusa-tsubota-241485358 | https://github.com/tsubokazu

Summary

Senior AI platform and infrastructure engineer targeting Lead ML Architect roles for Google Cloud. Designs and operates production-grade ML and MLOps platforms spanning Vertex AI, Gemini Enterprise Agent Platform Pipelines, GKE, Cloud Composer, Dataflow, BigQuery, IAM, and Terraform. Delivers end-to-end ML lifecycle architectures with reusable pipeline components, controlled promotion, model registry and lineage practices, monitoring, retraining, rollback, and reproducibility standards. Brings hands-on architectural decision-making across managed and Kubernetes-based serving models, cloud security, governance, observability, and scalable GPU-backed AI workloads.

Experience

Senior AI Platform & MLOps Infrastructure Engineer — Mentanomaly
Jul 2024 – Present

- Designed an end-to-end GCP ML platform architecture using Vertex AI for ML lifecycle management and Gemini Enterprise Agent Platform Pipelines for controlled, agent-driven workflow execution.
- Built modular, reusable pipeline components with defined inputs and outputs, separating AI orchestration from compute-intensive inference so stages could be independently developed, retried, scaled, and replaced.
- Built and operated production ML workloads across Vertex AI, GKE, Cloud Composer, Dataflow, Cloud Storage, IAM, and Terraform; evaluated Vertex AI versus Kubernetes serving through scalability, control, latency, and operational-complexity trade-offs.
- Established BigQuery as a centralized analytical layer, transforming raw data into curated training datasets and reusable features with schema consistency, partitioning, access control, data quality, and workload isolation.
- Designed ML lifecycle governance covering feature engineering, evaluation, model registration and lineage, approval-based promotion, CI/CD validation, retraining, rollback mechanisms, artifact retirement, and reproducibility across development, testing, and production environments; reduced model release time from 20 to 14 minutes.
- Implemented Monitoring & Observability for model quality, data and model drift, ground-truth validation, inference latency, failures, and GPU utilization using Prometheus, Grafana, OpenTelemetry, and tracing; reduced mean time to detect and recover from production ML issues from 2 hours to 20 minutes.
- Designed and operated a GPU-backed, asynchronous image-generation platform supporting over 10K users and 30K daily generation jobs; improved inference throughput by 8%, reduced p95 generation latency from 2.5 seconds to 1.7 seconds,

increased GPU utilization from 83% to 92%, and reduced infrastructure cost per generation by 7%.

- Applied least-privilege IAM, service identities, environment isolation, network boundaries, secret-management, auditability, and resource ownership controls, balancing cloud security and governance against scalability, performance, operability, and cost.

Senior AI Production & Cloud Infrastructure Engineer — ScalyX.ai
Jul 2022 – Jul 2024

- Designed production AI-serving architecture across GCP and AWS, using Vertex AI, GKE, Cloud Composer, Dataflow, Cloud Storage, IAM, Terraform, and Kubernetes-based serving to support scalable ML workflows.
- Developed reusable, asynchronous pipeline stages for AI orchestration, model inference, post-processing, retries, and failure recovery; enabled independent scaling of CPU- and GPU-bound components and recovery without restarting complete workflows.
- Built a resilient RAG and tool-calling pipeline using Python, FastAPI, PostgreSQL with pgvector, and Redis, covering ingestion, embeddings, retrieval, generation, orchestration, memory, caching, retries, timeouts, and fallbacks.
- Implemented monitoring for model quality, data and model drift, inference latency, failures, GPU utilization, LLM cost, and workflow performance with Prometheus, OpenTelemetry, and Grafana; reduced mean time to detect and recover from ML production issues from 2 hours to 20 minutes.
- Managed model deployment and versioning, IAM controls, Cloud Storage integration, Terraform provisioning, and CI/CD-oriented testing for AI services, improving reliability across evolving multi-tenant retail workflows.

Senior Backend & Cloud Infrastructure Engineer — Madoromi, Inc.
Jul 2018 – Jul 2022

- Progressed from junior engineering to senior-level ownership across SaaS and AI-powered applications, designing APIs and data models, resolving production issues, optimizing performance, and contributing to backend architecture.
- Expanded responsibility into AWS cloud infrastructure, containers, CI/CD, monitoring, networking, databases, deployment automation, and production operations to strengthen reliability, scalability, and operational readiness.
- Facilitated technical discussions with customers and senior stakeholders, translating business requirements into API, data-flow, infrastructure, security, scalability, and observability designs.
- Evaluated and documented architecture trade-offs between managed and self-managed infrastructure, synchronous and asynchronous processing, and centralized versus independently scalable services, creating practical implementation plans aligned with stakeholder priorities.

Skills

Google Cloud ML Platform: Google Cloud Platform (GCP), Vertex AI, Gemini Enterprise Agent Platform Pipelines, BigQuery, GKE, Cloud Composer

MLOps & ML Lifecycle: Pipeline orchestration, Model Registry & lineage, CI/CD for ML, Model monitoring, drift detection & retraining

Cloud Architecture & Governance: IAM & least-privilege security, Terraform, Monitoring & Observability, Kubernetes

Education

Kobe University

Bachelor of Engineering, Computer and Intelligent Systems Engineering

2018